

Power BI Internship Project

Hospital Readmission & Patient Outcome Dashboard

Business Context

You are a **Junior Data Analyst Intern** working with a hospital management team. Your responsibility is to use **Power BI** to analyze hospital admission data and help management understand:

- 1 Patient readmission patterns
- 2 Department performance
- 3 Cost vs outcome efficiency

Data Source

- 1 File: **hospital_admissions.csv**
- 2 Import the cleaned dataset into Power BI
- 3 Perform additional transformations using Power Query where necessary

Part 1: Power Query (Data Preparation)

Perform the following data preparation steps:

- 1 Confirm correct data types for admission date, discharge date, treatment cost, and length of stay
- 2 Check for blank diagnosis values
- 3 Check for inconsistent department names
- 4 Create a new column: **Discharge Month** from discharge date
- 5 Create a conditional column: **Readmission Flag** (Yes = 1, No = 0)
- 6 Create a column **Age group** with range 0-18, 19-40, 41-60, 61+

Part 2: DAX Measures (Core Metrics)

- 1 Total Patients
- 2 Average Length of Stay
- 3 Average Treatment Cost
- 4 Overall 30-Day Readmission Rate

Part 3: Dashboard Pages & Visual Requirements

Page 1: Hospital Overview

- 1 KPI Cards: Total Patients, Average Length of Stay, Average Treatment Cost, Overall Readmission Rate
- 2 Bar Chart: Patients by Department

3 Slicers: Department, Discharge Month

Page 2: Readmission Analysis

1 Line Chart: Monthly 30-day readmission rate

2 Bar Chart: Readmission rate by department

3 Column Chart: Readmission rate by age group

4 Table: Department, Average Length of Stay, Readmission Rate

Page 3: Cost & Outcome Analysis

1 Pie or Donut Chart: Patient outcome distribution

2 Bar Chart: Average treatment cost by outcome

3 Scatter Chart: Length of stay vs treatment cost (by outcome)

4 Table: Department, Average Cost, Readmission Rate

Part 4: Analysis Questions (Power BI Insights)

- 1 Which month has the highest readmission rate?
- 2 Which department contributes most to readmissions?
- 3 Do shorter hospital stays result in higher readmissions?
- 4 Which age group is most at risk?
- 5 Are higher treatment costs linked to better patient outcomes?
- 6 Which departments are high cost but poor outcome?

Part 5: Management Summary

Create a one-page written summary that highlights key findings from the dashboard, identifies departments needing improvement, patient groups requiring follow-up care, and provides clear data-driven recommendations for management.